



2026 IAIFI Summer Workshop



Dropout Universality

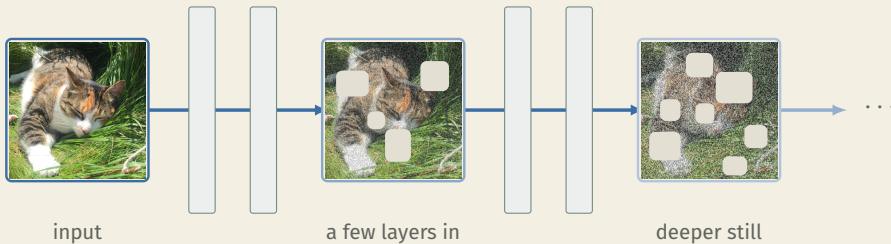
Critical Scaling and Optimal Scheduling at the Edge-of-Chaos

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A usual suspect of regularization: dropout



A fresh mask at every layer: the pattern changes and the damage compounds.

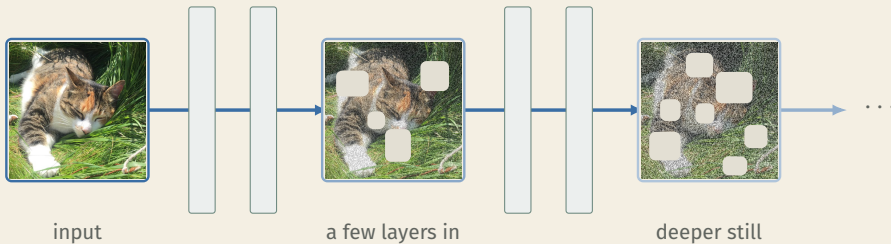
The standard theory

Usual NNs are overparametrized, so there are enough parameters to memorize training neural patterns ("coadaptation"). Dropout blocks activations randomly, mitigating this.

Srivastava et al., JMLR **15**, 1929 (2014).

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The practice

People usually do a grid search over the probability of dropping out and call it a day.

Reasonable. But is it **optimal**?
Can we do better?



Front-load your dropout!

TLDR: Front-load your dropout!



constant



layer ℓ

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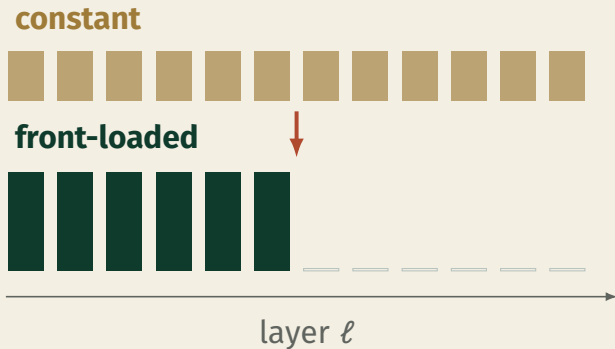


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constant



front-loaded



layer ℓ

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Final-epoch test loss

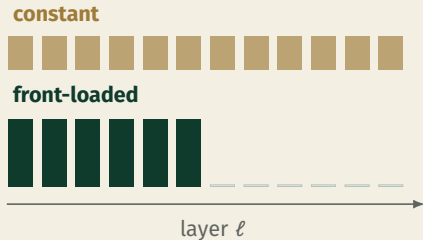
18–35% lower (MLP)

4–6% lower (ViT)

Epochs to reach constant-dropout accuracy

29–31 vs 75 (**$\approx 60\%$ fewer**)

TLDR: Front-load your dropout!



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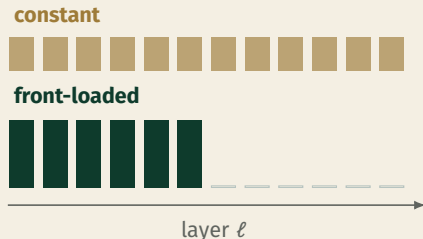
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TLDR: Front-load your dropout!



Same total dropout. Same cost per epoch.

Final-epoch test loss

18–35% lower (MLP) **4–6%** lower (ViT)

Epochs to reach constant-dropout accuracy

29–31 vs 75 (**~60% fewer**)

Follows from applying principle of tuning to **criticality** to dropout distribution.

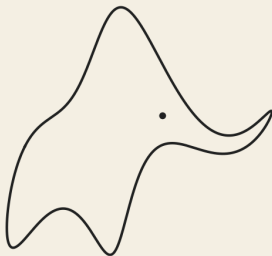
In search of better variables



“With four parameters I can fit an elephant, and with five I can make him wiggle his trunk.”

von Neumann to Fermi, via Dyson

Can a few **collective variables** describe what depth does to information?



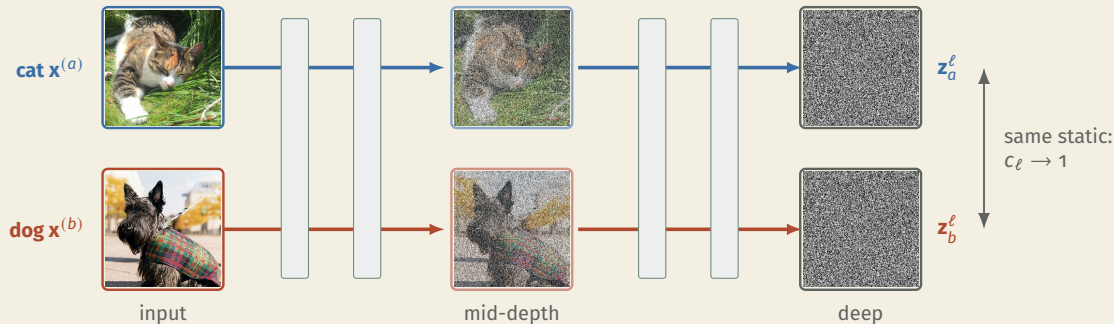
Mayer, Khairy & Howard,
Am. J. Phys. **78**, 648 (2010)

At initialization, depth is like broken telephone



Each layer *re-transmits* the message it receives. Track the correlation between the two inputs, c_ℓ .

Aim of the game: postpone perfect correlation for as long as possible!



Initialization sets how long distinctions survive



Ordered

$c_\ell \rightarrow 1$ fast;
everything looks alike

ordered



Critical edge

forgetting is slow;
distinctions survive deep

edge-of-chaos



Chaotic

differences blow up,
then saturate

chaotic



layer 0

A circular input manifold through random tanh layers, projected to 2D.

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Tune initialization to the edge so distinctions propagate deepest.

The edge of chaos is a borrowed hypothesis



The critical brain hypothesis

Cortical activity arrives in avalanches whose sizes follow a power law, $P(s) \sim s^{-3/2}$. Scale-free all the way up to the recording array — electrodes $200\ \mu\text{m}$ apart, millimetres of cortex firing as one event. No intrinsic scale, so the correlation length $\xi \rightarrow \infty$.

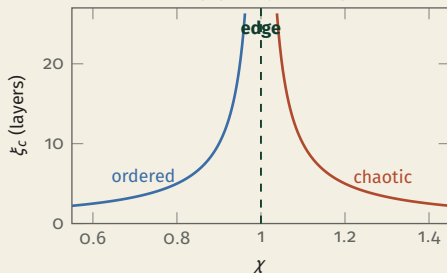
Beggs & Plenz, J. Neurosci. **23**, 11167 (2003). Still contested:
Beggs & Timme, Front. Physiol. **3**, 163 (2012).

Inherited by machine learning

tuning nets to edge-of-chaos at initialization

Sompolinsky et al. 1988; Packard 1988; Bertschinger & Natschläger 2004; Poole et al. 2016; Schoenholz et al. 2017.

The smoking gun: ξ diverges



Criticality's signature is a **diverging correlation length**
 $\xi \rightarrow \infty$ and power law behaviour.

In a deep network that length is measured in *layers*.

Anything that cuts off the divergence destroys criticality ...**enter dropout**

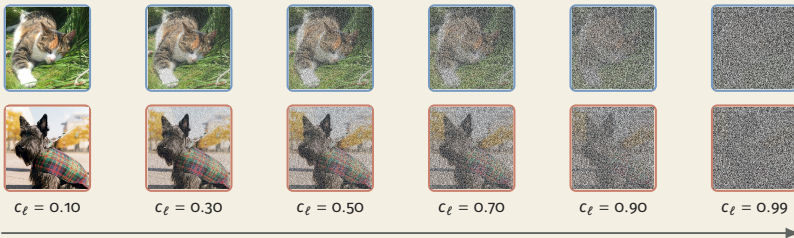
Distance to perfect correlation decays exponentially



How far are we from perfect correlation at layer ℓ ? Track $|1 - c_\ell|$: it decays exponentially, at a scale mean-field theory hands us explicitly:

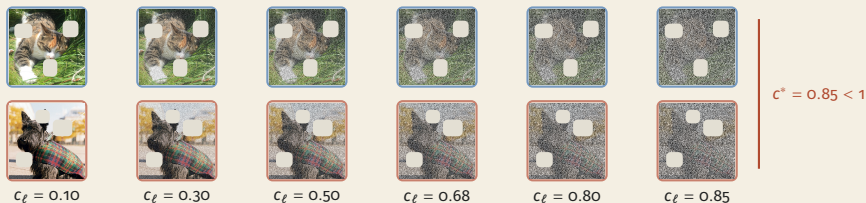
$$|1 - c_\ell| \sim e^{-\ell/\xi_c}, \quad \xi_c^{-1} = -\log \left[\sigma_w^2 \int Dz_1 Dz_2 \phi'(u_1^*) \phi'(u_2^*) \right].$$

σ_w the weight std, ϕ the nonlinearity, $u_{1,2}^*$ the two inputs' preactivations at the fixed point. So ξ_c is set by σ_w, σ_b, ϕ alone — knobs we tune to make it **diverge**.



Tune to the edge and $\xi_c \rightarrow \infty$: the two never collapse together.

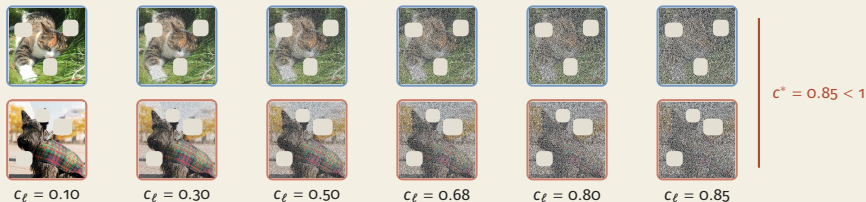
Dropout is an external field



Each row gets its own mask, so even in pure noise the two never agree: c_ℓ stalls below 1.

$c = 1$ **stops being a fixed point**, and ξ_c was a linearization *about* it — so it is not even defined. Look for the fixed point *perturbatively away* from perfect correlation, in the dropout field $h \propto 1 - \rho$.

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Tuned to the edge ($t \equiv \chi_\rho - 1 = 0$), one can show a length comes back:

$$\xi_c \propto h^{-1/2}.$$

Two universality classes, neither of them Ising



The critical exponents are set by the **analyticity of the activation**, not by a spatial dimension.

	β	δ
Smooth ϕ (tanh)	1	2
Kinked ϕ (ReLU)	2	$\frac{3}{2}$
Ising, Landau mean field	$\frac{1}{2}$	3

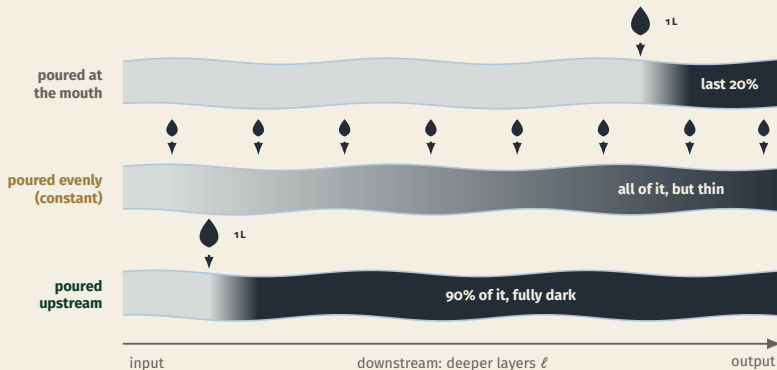
A kink in ϕ puts a branch point in $\bar{F}_\rho$ at $c = 1$: $m^{5/2}$ replaces m^3 .

Structural analogy only: depth is an iteration count, not a distance, and $m \geq 0$ carries no $\mathbb{Z}_2$ symmetry.

Intuition: where would you pour the ink?



Same litre of ink, same river, three ways to pour it in.



Dropout is the ink, depth is the river. Constant dropout is the middle one — front-loading is the bottom one.

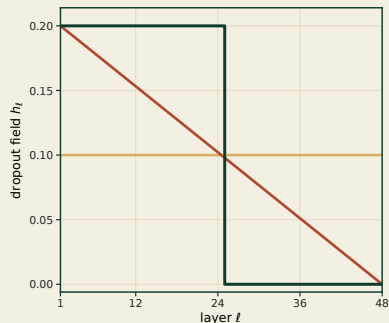
Optimal schedule: saturated blocks, placed early



Depth acts like an RG scale, so the couplings *run*: the dropout field becomes a function of depth, $h \rightarrow h_\ell$.

Treat the schedule as a functional and maximize the correlation length at fixed budget:

$$\max_{\{h_\ell\}} \xi_{\text{eff}}[h] \quad \text{s.t.} \quad \frac{1}{L} \sum_{\ell} h_\ell = \bar{h}, \quad 0 \leq h_\ell \leq h_{\text{max}}.$$



— constant — linear front — step front

Three profiles, identical $\sum_{\ell} h_\ell$.

Optimal schedule: saturated blocks, placed early

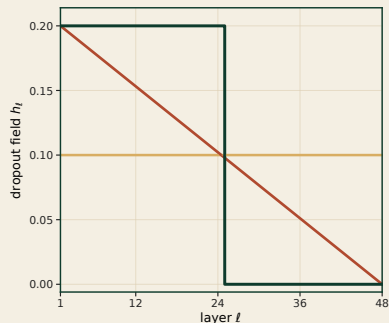


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At the edge $\xi_{\text{eff}}^{-1}[h] \propto \frac{1}{L} \sum_{\ell} h_\ell^a$ with $a < 1$, so maximizing ξ_{eff} means minimizing a **concave** sum at fixed mean. The optimum sits at the corners: $h_\ell \in \{0, h_{\text{max}}\}$.



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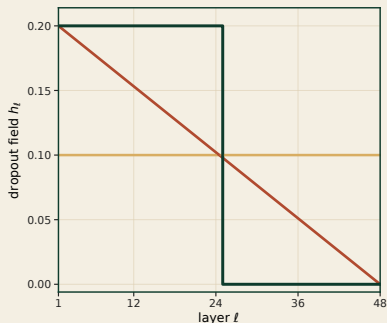
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Propagation fixes the shape. It is order-invariant, so it cannot say *where* the block goes.

Reach fixes the position. An early mask perturbs a longer downstream suffix.

Reach is a separate argument, and the weaker of the two.



— constant — linear front — step front

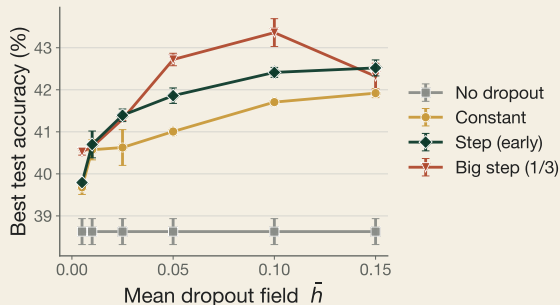
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The predicted schedule wins, and wins sooner



Best test accuracy across schedules

CIFAR-10, ReLU MLP, width 256



Mean $\pm$ s.e. over 20 runs.

Left panel is accuracy; the percentages are cross-entropy. “Fewer epochs” is epoch-proportional compute at identical architecture and batch size, not wall-clock or per-step FLOPs.

Final-epoch test loss

18–35% lower

MLP settings

4–6% lower

Vision Transformers

Epochs to match constant dropout

29–31 vs 75

≈60% fewer, at equal per-epoch cost



From microscopic masks to a macroscopic design rule

Dropout hurts propagation.



Maximize the correlation length.



Front-load dropout.

Outlook: treat other depth-dependent hyperparameters as dynamical fields.

Questions?

arXiv:2605.21648 · Carnegie Mellon University

Backup: wide networks collapse to one map in depth



At $N \rightarrow \infty$ the layer- ℓ theory is *free*: the action is quadratic, so the propagator q_{ab}^ℓ is its only datum.

$$p(z^\ell) \propto e^{-S_\ell[z]}, \quad S_\ell = \frac{1}{2} \sum_{i,a,b} (q_\ell^{-1})_{ab} z_{i;a} z_{i;b}$$

Integrating out a layer makes it *flow*:

$$q_{ab}^{\ell+1} = \sigma_w^2 \langle \phi(z_a) \phi(z_b) \rangle_{q_\ell} + \sigma_b^2.$$

Once $q_{aa}^\ell \rightarrow q^*$, only $c_\ell = q_{ab}^\ell / q^*$ survives:

$$c_{\ell+1} = F(c_\ell)$$

$$F(c) = [\sigma_w^2 \langle \phi(u_a) \phi(u_b) \rangle_c + \sigma_b^2] / q^*$$

(u_a, u_b) Gaussian, $\mathbb{E}[u_a u_b] = c q^*$.

Millions of weights become a one-dimensional dynamical system in depth.

The quadratic action is exact only at $N = \infty$; finite width generates interactions at $O(1/N)$. Without dropout $\lambda = F'(1) \equiv \chi$.

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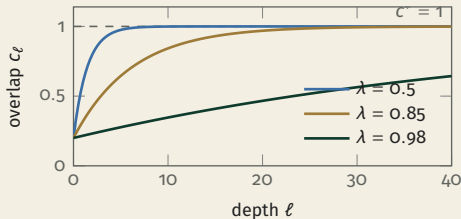
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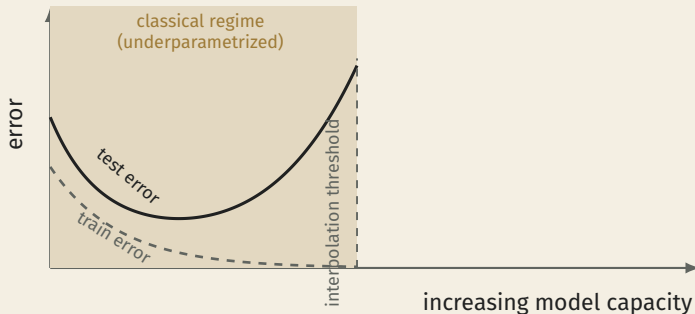
Same fixed point $c^* = F(c^*)$; only the rate differs.

The slope $\lambda = F'(c^*)$ sets an e-folding depth

$$\xi_c = -1/\log |\lambda| \quad \text{layers.}$$

Here 1.4, 6, 50.

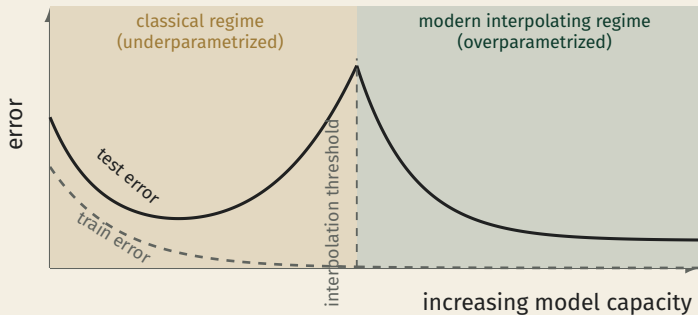
Backup: regularization lost its textbook justification



Classical regime

More capacity overfits, so a regularizer earns its keep by *removing* capacity.

Backup: regularization lost its textbook justification



Classical regime

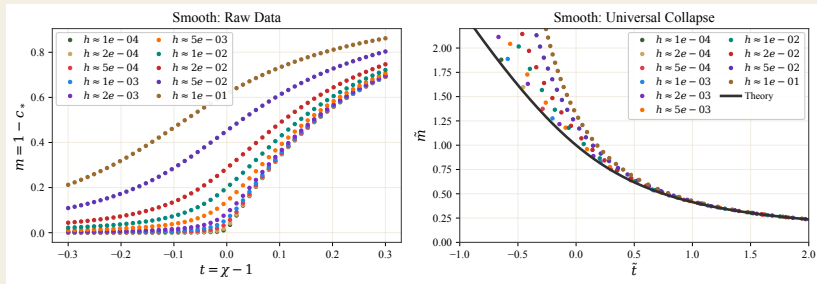
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Modern interpolating regime

Past the threshold more capacity *helps*. Our networks live here, where that story fails.

So what *is* dropout doing, how much of it, and **where in depth?**

Backup: two parameters collapse onto one scaling variable



$$\text{smooth } m = h^{1/2} \mathcal{M}_s(t/h^{1/2}) \quad \cdot \quad \text{kinked } m = h^{2/3} \mathcal{M}_k(t/h^{1/3})$$

Solid line is the parameter-free normal form $\tilde{m} = \sqrt{1 + \tilde{t}^2} - \tilde{t}$, not a fit.

Collapse holds in the scaling region; the spread at $\tilde{t} \lesssim -0.5$ is corrections to scaling at the largest h .

Backup: the theory is at initialization; the results are trained



The gap, stated plainly. ξ_{eff} is a property of the *forward map at initialization*. The experiments measure test accuracy after full training. Nothing here derives the second from the first.

What is actually claimed

- ▶ The schedule shape is chosen *a priori* from the initialization theory, with no tuning on the test set.
- ▶ That prediction, made in advance, is the one that wins.
- ▶ So the initialization functional is a useful *design heuristic* for a schedule, not a theorem about the trained network.

Why it is not vacuous. The ordering of schedules is predicted before any training run, and the predicted ordering is reproduced across MLPs and ViTs, two architectures with different F .

Honest answer if pressed: connecting init-time ξ_c to end-of-training generalization is open, and is the main thing I would want to do next.

Backup: “isn’t this all just mean field?”



Yes, and that is not the interesting axis.

- ▶ $N \rightarrow \infty$ is exactly a mean-field limit. There are no fluctuation corrections at leading order.
- ▶ There is **no spatial dimension** here. Depth is an iteration index, not a coordinate, so there is no d , no upper critical dimension, and no ϵ -expansion.
- ▶ The universality is therefore not over dimension. It is over the **analyticity class of ϕ** at the fixed point.

Why the exponents still differ from Landau. A kink in ϕ puts a branch point in $\bar{F}_\rho$ at $c = 1$. The effective potential picks up $m^{5/2}$ rather than m^3 , and the exponents follow from that non-analyticity. Same mean-field machinery, different local expansion.

So “mean field” describes the method; it does not determine the class.



- ▶ Finite N makes c_ℓ a random variable with $O(1/N)$ variance about the deterministic recursion.
- ▶ Near the edge, $\lambda \rightarrow 1$, so fluctuations accumulate over depth rather than damping. The width needed for the mean-field description to hold grows with L .
- ▶ Experiments run at width 256 with L of order tens, where the recursion tracks the empirical overlap well.
- ▶ The scheduling conclusion depends only on *concavity* of $h \mapsto h^a$, which is robust to the fluctuation correction shifting a slightly.

The $1/N$ corrections are not computed in this work. Roberts, Yaida & Hanin (2022) develop the systematic expansion.

Backup: why the ViT numbers transfer



The fair objection. Attention, LayerNorm and residual connections all change F . The MLP derivation does not literally apply.

What survives the change of F

- ▶ The argument needs only two things: dropout acts as a decorrelating field $h > 0$, and ξ_{eff}^{-1} is *concave* in h .
- ▶ Neither depends on the specific form of F , only on the existence of a fixed point at $c = 1$ that dropout displaces.
- ▶ So the prediction is qualitative for ViTs: front-load, whatever the exponent a turns out to be.

Observed. 4–6% final-epoch loss reduction, same ordering of schedules. Component ablations put dropout on the attention block, the MLP block, and both; the early-step schedule wins in every case.

The ViT exponent is not measured. Only the sign of the effect is predicted.

Backup: the full second-moment recursions



$$z_{i;a}^{\ell+1} = \sum_{j=1}^N W_{ij}^{\ell+1} \phi(z_{j;a}^{\ell}) + b_i^{\ell+1},$$
$$W_{ij}^{\ell} \sim \mathcal{N}(0, \sigma_w^2/N), \quad b_i^{\ell} \sim \mathcal{N}(0, \sigma_b^2).$$

$$q_{aa}^{\ell+1} = \sigma_w^2 \int Dz \phi^2 \left(\sqrt{q_{aa}^{\ell}} z \right) + \sigma_b^2,$$
$$q_{ab}^{\ell+1} = \sigma_w^2 \mathbb{E}_{(u_a, u_b)} [\phi(u_a) \phi(u_b)] + \sigma_b^2,$$
$$c_{ab}^{\ell} = q_{ab}^{\ell} / \sqrt{q_{aa}^{\ell} q_{bb}^{\ell}}.$$

Assumptions

- ▶ $N \rightarrow \infty$
- ▶ i.i.d. Gaussian init
- ▶ fixed activation ϕ
- ▶ exchangeable neurons

(u_a, u_b) jointly Gaussian with covariance fixed by $q_{aa}^{\ell}, q_{bb}^{\ell}, q_{ab}^{\ell}$; Dz is the standard Gaussian measure.

At equal variance,

$$\mathbb{E} \|\mathbf{z}_a - \mathbf{z}_b\|^2 / N = 2q^*(1 - c).$$

Backup: depth as an RG scale



network depth $\ell \uparrow \longleftrightarrow$ QFT scale $\mu \downarrow$ toward the infrared

Coarse-graining representations makes the effective coefficients run with depth:

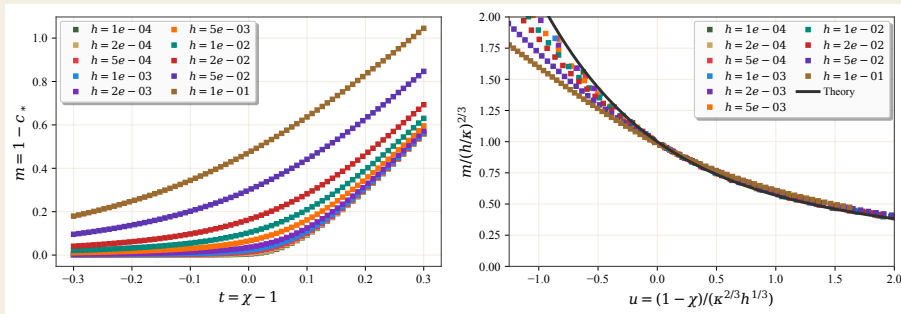
coefficient	network definition	effective-theory role
t_ℓ	$\chi_{\rho_\ell} - 1$	mass-like detuning
h_ℓ	$1 - \bar{F}_{\rho_\ell}(1)$	relevant decorrelation source
g_ℓ	$\bar{F}_{\rho_\ell}''(1)$	smooth analytic interaction
κ_ℓ	branch-point amplitude	kinked non-analytic interaction

Both equations of state follow from a reduced (zero-momentum) 1PI action

$$\Gamma_{\text{eff}} = -\frac{t}{2}m^2 + \mathcal{V}(m) - hm, \quad \mathcal{V}_s = \frac{g_\rho}{6}m^3, \quad \mathcal{V}_k = \frac{2\kappa}{5}m^{5/2}.$$

RG-scale analogy. Generic MLPs are not thereby proven conformal, and Γ_{eff} is not the training loss or a neural partition function. Roberts, Yaida & Hanin (2022), Ch. 4.

Backup: the kinked (ReLU) collapse



$m_k(t, h) = h^{2/3} \mathcal{M}_k(t/h^{1/3})$, with $\mathcal{V}_k = \frac{2\kappa}{5} m^{5/2}$ from the branch point at $c = 1$.

Ordinary bias-free ReLU follows the constrained path $t = -h$ and retains $\xi \sim h^{-1/3}$.

Backup: the complete critical-exponent dictionary



Universality class	ν_t	β	θ_{rel}	γ	δ	ν_ρ	α
Smooth activation	1	1	1	1	2	$\frac{1}{2}$	-1
Kinked activation	1	2	2	1	$\frac{3}{2}$	$\frac{1}{3}$	-3
Spin glass (SK MF)	$\frac{1}{2}$	1	2	1	2	$\frac{1}{4}$	-1
Ising-like (Landau MF)	$\frac{1}{2}$	$\frac{1}{2}$	1	1	3	$\frac{1}{3}$	0

$$m \sim |t|^\beta \quad (h = 0),$$

$$m \sim |h|^{1/\delta} \quad (t = 0),$$

$$\chi_M \sim |t|^{-\gamma},$$

$$C \sim |t|^{-\alpha}.$$

$$\xi \sim |t|^{-\nu_t} \quad (h = 0),$$

$$\xi \sim |h|^{-\nu_\rho} \quad (t = 0),$$

$$m_\ell \sim \ell^{-\theta_{\text{rel}}} \quad (t = h = 0).$$

The kinked ν_ρ is a normal-form field exponent.

Backup: relation to prior scheduled-dropout work



- ▶ **Stochastic depth** (Huang et al., 2016) drops whole residual blocks with a probability that increases with depth. A mechanism, not a derived profile.
- ▶ **Curriculum dropout** (Morerio et al., 2017) anneals dropout over *training time*. Orthogonal axis: ours is a profile over *layers*.
- ▶ **LayerDrop** (Fan et al., 2019) removes transformer layers for inference efficiency.
- ▶ **Hayou et al. (2019)** observed that smooth and ReLU-like activations behave differently at the edge of chaos.

What is new here. The exponents are extracted and the two universality classes identified, and the schedule shape is *derived* from concavity of the correlation-depth functional rather than proposed and then tuned.